Advanced Topics in Computer Graphics II

Universität Bonn
Institut für Informatik II
03.11.2025

Winter term 2025
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Sheet G04 - Meshes Representations

Solutions for the theoretical and practical part via eCampus by Mo, 10.11.2025, **10:00**.

Practical Part

Assignment 1) Boundary Operator *(4Pts)*

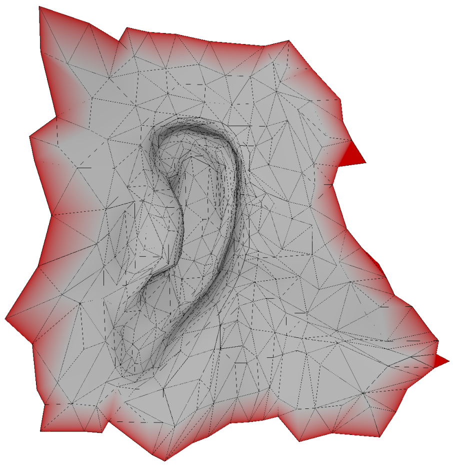

Figure 1: Result of applying the boundary operator on the mesh.

Download `framework_g04.zip` from eCampus and add it into the project. Your task is to detect the boundary of a mesh using the boundary operator from the lecture.

a) In `compute_boundary_operator(...)`, construct the boundary operator as a sparse matrix. For the boundary operator, you have to define an arbitrary (but fixed) order of the primitives. Think about the orientation of half-edges to define a well-defined ordering.

b) In `compute_boundary_edges(...)`, find the boundary edges using the boundary operator.

c) In `color_mesh(...)`, find all boundary edges and set the color of the corresponding vertices to red.

If you implemented everything correct, your output should look like Figure 1.
Hint: It is advised to not use the C++ `auto` keyword when using types from Eigen (the linear algebra library we are using). Eigen relies on compile time optimizations and if types are not hard defined, it may perform implicit casting which can lead to inefficient code.

Hint: Do not use the OpenMesh functions to detect boundary edges. Only use information about the orientation to construct the boundary operator.

Submit this exercise by uploading the complete source file onto eCampus.

Theoretical Part

Assignment 2) Halfedge Structure *(2Pts)*

In the half-edge structure, a vertex has only a pointer to an arbitrary outgoing half-edge. Describe how one can find the faces adjacent to a given vertex.

Assignment 3) Boundary Operator *(5Pts)*

A 2D mesh is given by its vertex and triangle lists

$$V := \begin{cases} v_0 = & (0.0, 0.0) \\ v_1 = & (-1.5, 3.0) \\ v_2 = & (-1.5, -0.5) \\ v_3 = & (0.0, -2.0) \\ v_4 = & (1.5, -0.5) \\ v_5 = & (1.5, 3.0) \end{cases}$$

$$F := \begin{cases} f_0 = & (v_0, v_1, v_2) \\ f_1 = & (v_0, v_2, v_3) \\ f_2 = & (v_0, v_3, v_4) \\ f_3 = & (v_0, v_4, v_5) \\ f_4 = & (v_0, v_5, v_1) \end{cases}$$

a) Calculate the 0-, 1-, and 2-simplex lists K^0, K^1, K^2 of the mesh. Assume that all triangles have a positive orientation and explain how you chose the orientation of the other simplices.

b) Draw the mesh with directed edges (1-simplices) and oriented triangles (2-simplices) and label the simplices.

c) Build the boundary operator matrices ∂_1 and ∂_2 for the mesh using your simplex lists.

d) $c^2 = (1, 1, 1, 0, 0)$ is an indicator vector for the triangles in the mesh. Use the boundary operator ∂_2 to calculate the indicator vector of the boundary edges of c^2. Mark the corresponding the triangles and edges in your drawing.

Good luck!